

DTI FORULA STUDENT MOTOR

Specification for F-MOT motor

Version 1.2

Valid for F-MOT-A inverter

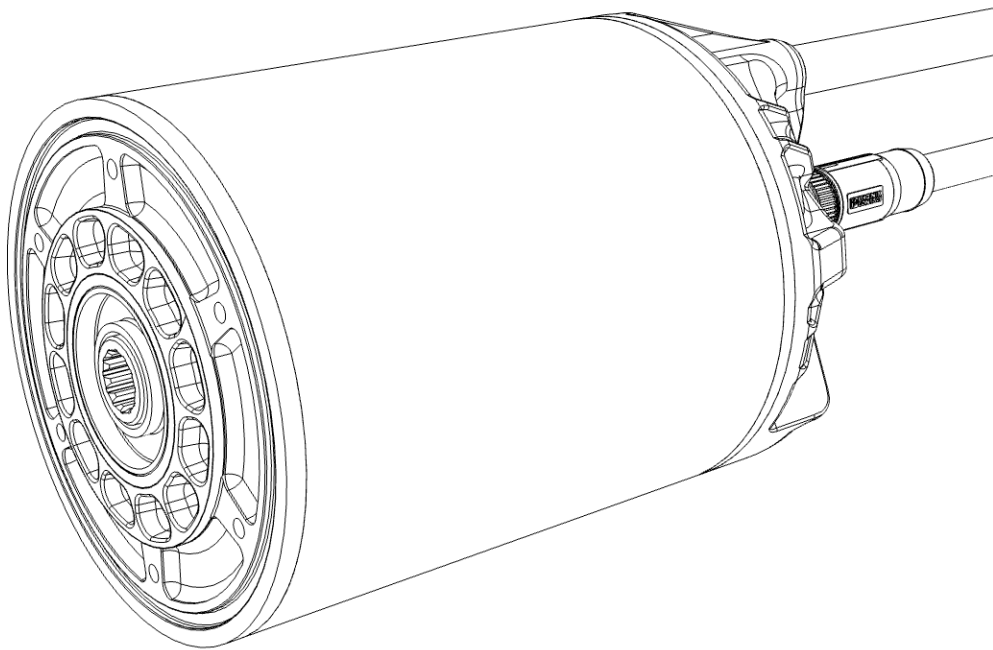


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1. HISTORY AND RELATED DOCUMENTS

1.1 Document history

10/2024	V1.0 Document created; basic specifications added
02/2025	V1.1 Aluminium material changed, additional appendixes remarked
02/2025	V1.2 Motor Torque-RPM chart added

1.2 Related documents

- F-MOT-A Motor Application Manual
- F-MOT-A Drawing_Cooling
- F-MOT-A Drawing
- F-MOT-A CAD Model
- F-MOT-A Shaft Model
- F-MOT-A Power Losses excel sheet

2. TECHNICAL SPECIFICATIONS

Parameters provided by Fischer Elektromotoren.

2.1 Motor peak and nominal condition

Table 1. motor peak and nominal specifications

	F-MOT-A	Unit
Peak Torque	31,6	Nm
Peak Current	66	Arms
Maximum Speed	20.000	rpm
Peak Power ¹	35	kW
Nominal Torque	24,6	Nm
Nominal Current	22,6	Arms
Nominal Speed	13250	rpm
Nominal Power	15,4	Kw

2.2 Motor basic parameters

Table 2. basic specifications

	F-MOT-A	Unit
Torque Constant	0,492	Nm/Arms
BEMF Constant	0,031	Vrms/rpm
Maximum DC bus voltage	600	Vdc
Pole pairs	4	
Temperature sensor ²	PT1000	
Position sensor type	TE connectivity 2-1393047-5	
Peak efficiency	>96	%
Phase resistance at 20 °C ³	0,126	Ω
Phase inductance ⁴	0,393	mH

¹ Measured at 10.500 rpm

² Isolation voltage 2kV

³ Measured resistance per phase

⁴ Measured inductance per phase

2.3 Motor mechanical parameters

Table 3. mechanical specifications

	F-MOT-A	Unit
Weight	3,7	Kg
Diameter	94,5	mm
Length	113	mm
Length with connector housing	160	mm
Motor housing material	Aluminum Alloy 6082	
Motor shaft material	Titanium Grade 5 (Ti-6Al-4V)	
Motor shaft spline DIN 5480-30deg Flat root	Side fit-11x0,8x30°x12 9H 7h	

3. THERMAL AND MECHANICAL SPECIFICATION

To ensure that if the inverter is capable of current control (as in the case of the F-SIC inverter), but the user still needs to specify a torque value, a table is provided here to determine the correct values. Data provided by Fischer Elektromotoren.

Table 4. Motor torque and AC current

Motor AC current	Motor torque
5 Arms	2,6 Nm
11 Arms	5,6 Nm
16 Arms	8,2 Nm
21 Arms	10,6 Nm
26 Arms	13,1 Nm
31 Arms	15,5 Nm
36 Arms	17,9 Nm
41 Arms	20,4 Nm
46 Arms	22,7 Nm
51 Arms	25,0 Nm
56 Arms	27,3 Nm
61 Arms	29,5 Nm
66 Arms	31,6 Nm

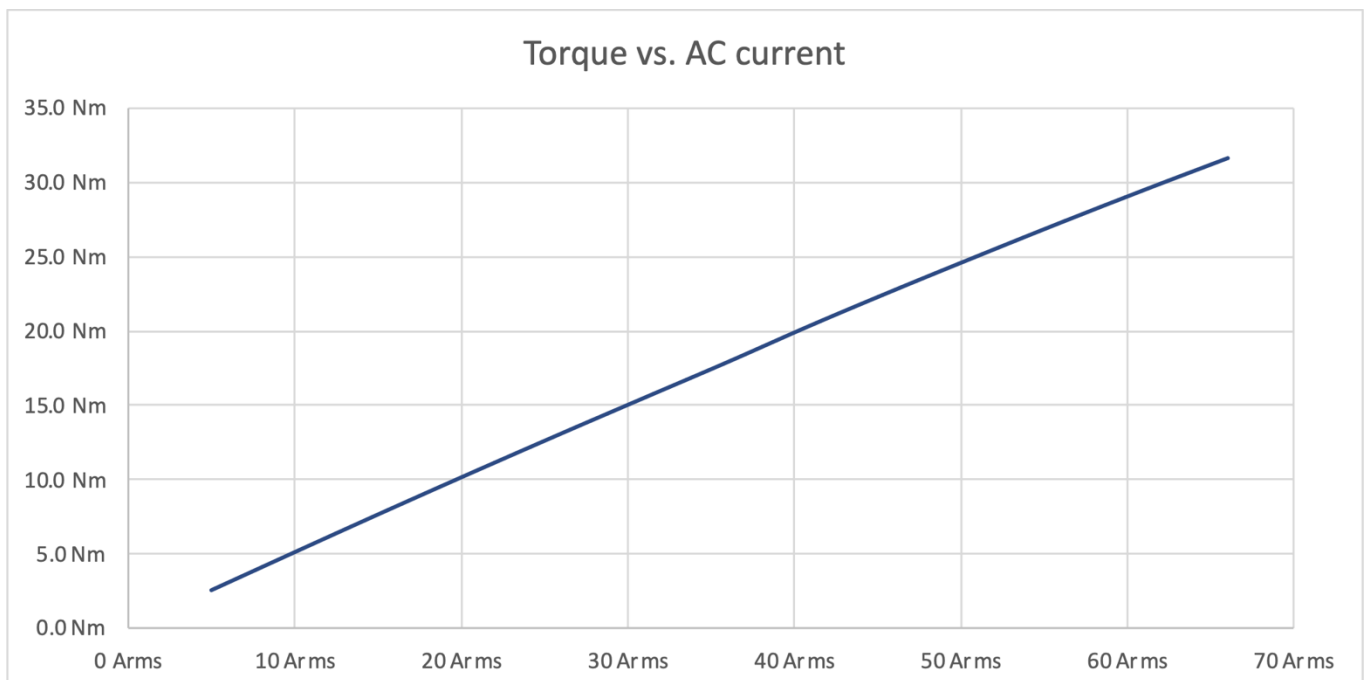


Figure 1 Motor torque and AC current

4. MOTOR POWER LOSSES

The motor generates two main types of losses. The first is the conductive loss, which arises in the motor stator windings and is dependent on the AC current. The second is the iron loss, which occurs in the core of both the stator and rotor. This loss is dependent on the rotational speed of the motor. The iron loss data was provided by Fischer Elektromotoren.

4.1 Total iron loss

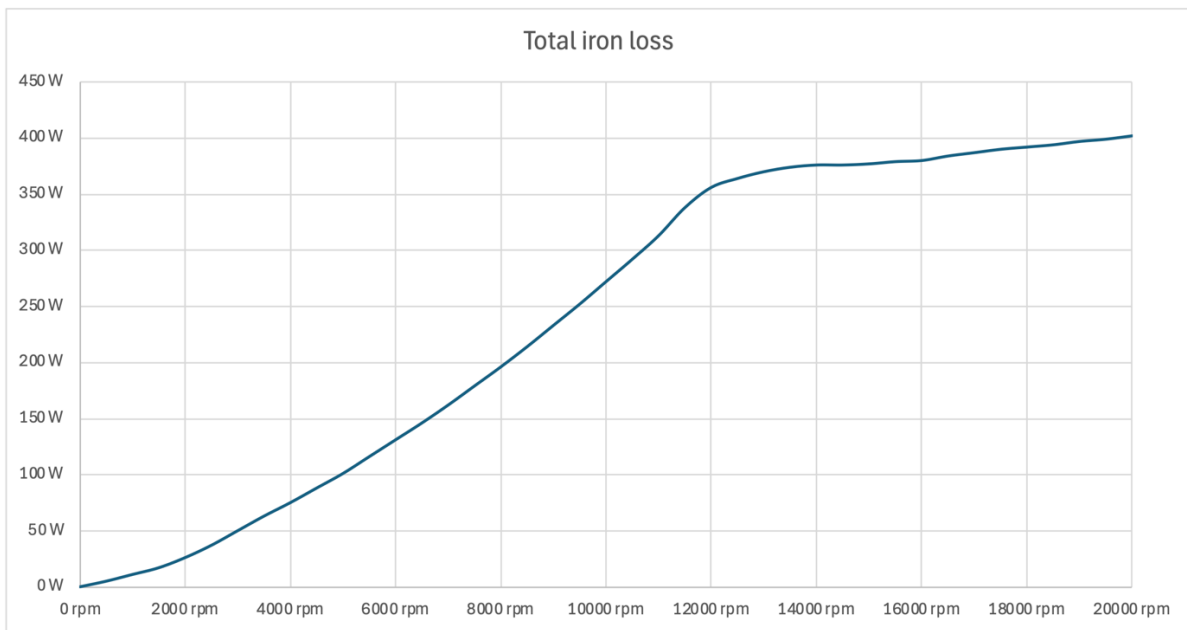


Figure 2 Total iron losses

4.2 Conductive loss

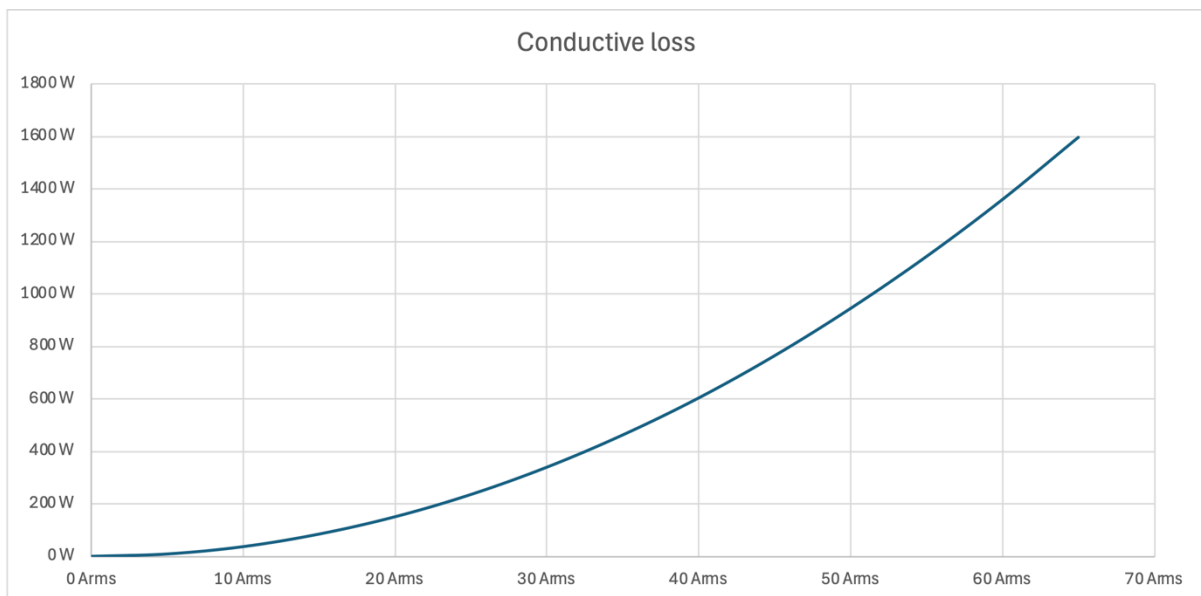


Figure 3 Conductive loss

4.3 Motor efficiency

The efficiency data was provided by **Fischer Elektromotoren**.

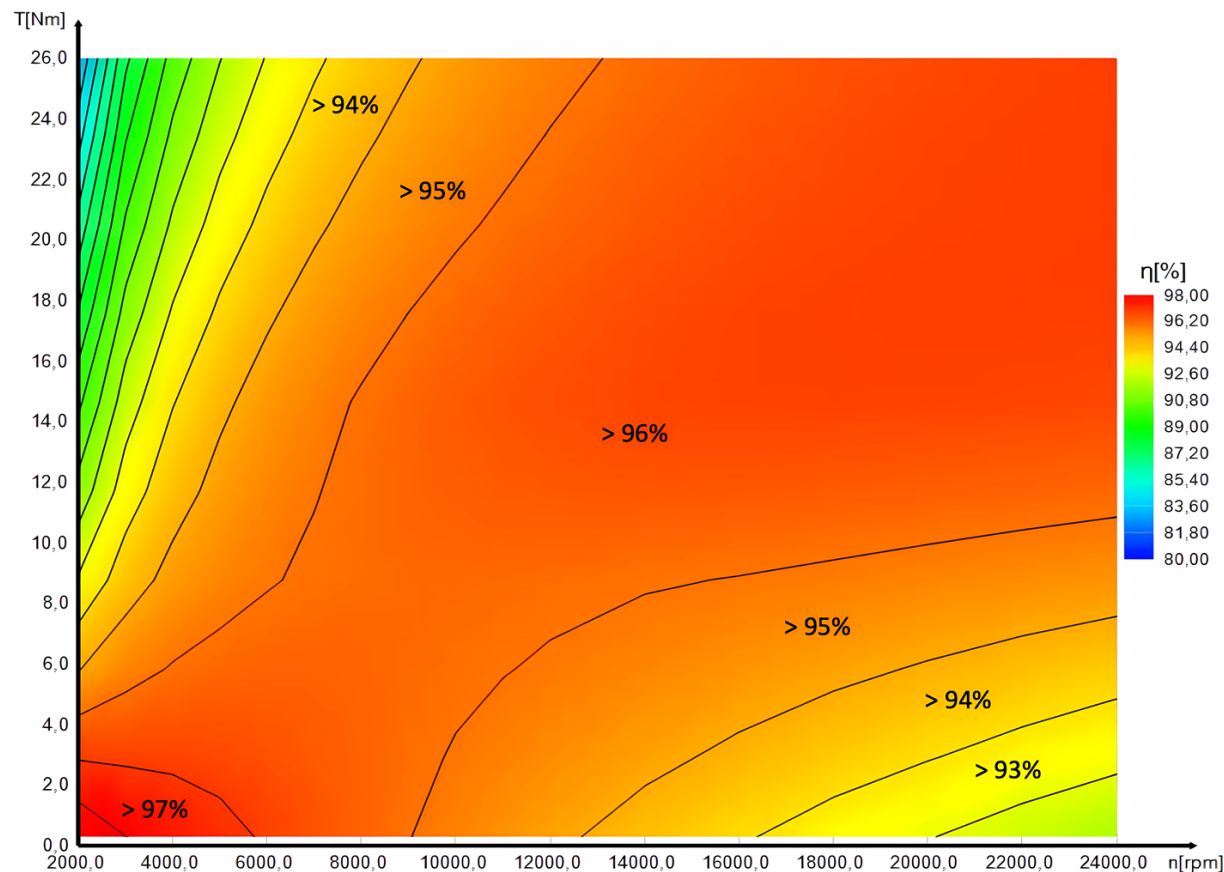


Figure 4 Motor efficiency graph

4.4 Torque vs. RPM

Data provided by Fischer Elektromotoren.

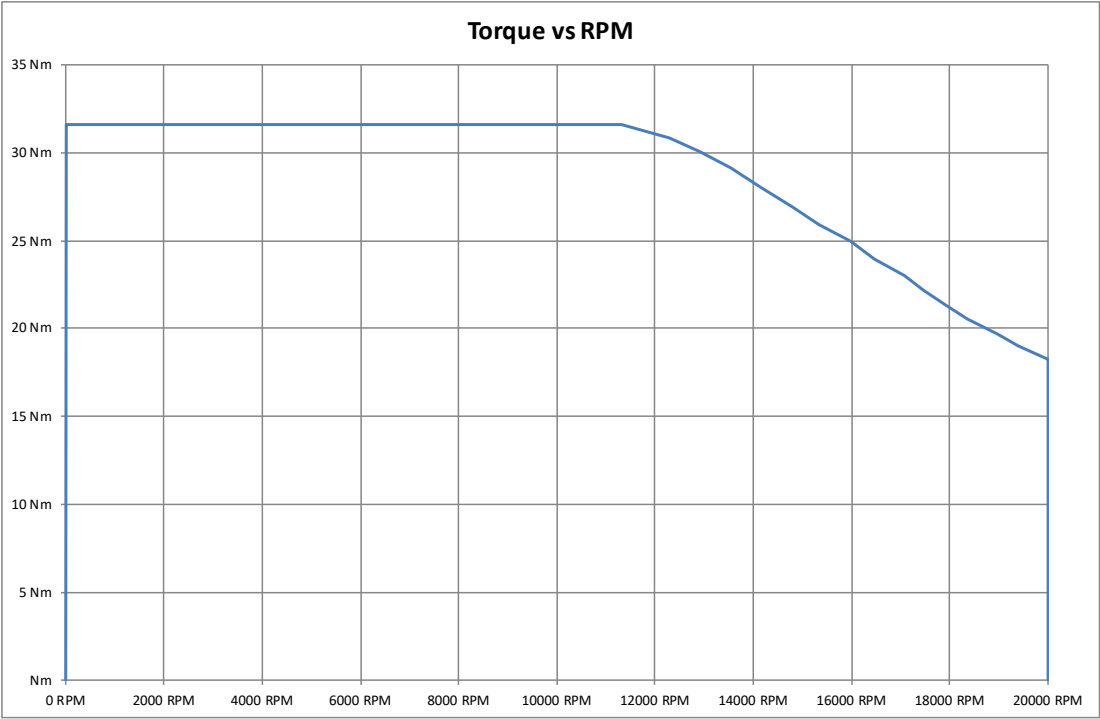


Figure 5 Torque vs. RPM Graph

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